Arwa Elsayed Eldesawy

Ai Engineer

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Profile

Entry-level AI Engineer with a strong foundation in **Machine Learning, Deep Learning, Computer Vision, and Generative AI**. Experienced in developing AI applications and intelligent workflows using **Python, PyTorch, TensorFlow, FastAPI, LLM APIs, and n8n**, with internship experience in **ML/DL model development and RAG**. Strong foundation in **AI integration, REST API development, and workflow automation**.

Education

Bachelor of Computer Science major: Artificial Intelligence, *Misr university for science and technology(MUST)* 2022 – 2026 | Cairo,Egypt
Cumulative GPA: 3.66/4.0

Skills

Technical Skills

- **Languages:** Python, C++, SQL
- **Machine Learning :** PyTorch, TensorFlow, Scikit-learn, Pandas
- **Computer Vision:** OpenCV, YOLOv8, CNNs, OCR, Transfer Learning
- **LLMs & Generative AI:** LLMs, Prompt Engineering, RAG, LLM APIs
- **AI Automation:** n8n, Workflow Automation
- **Databases:** PostgreSQL, MySQL, SQL Server, Oracle
- **Backend & APIs:** FastAPI, REST APIs, SQLAlchemy
- **Tools:** Git/GitHub, Jira, Agile, Pull Requests, Code Review, Cursor, Claude Code

Experience

Tech Track Participant,Digitera Program

- **Applied requirements analysis and product planning** :BRD-to-PRD, user stories, scope definition, and acceptance criteria.
- **Used Jira and Agile workflows** for backlog management, sprint planning, and delivery tracking.,**Applied AI-assisted development and engineering workflows** using Cursor, Claude Code, MCP, Git/GitHub, Pull Requests, code reviews, debugging, and testing.

AI Intern ,Internship at ENPPI

- **Trained ML & Deep Learning models** , **Applied Generative AI & RAG frameworks** to build intelligent, context-aware retrieval systems. **Integrated LLMs with vector embeddings** to automate data extraction and improve search accuracy.

Internship at ABE (agricultural bank of Egypt).

- **Designed and queried relational databases** using SQL to clean, filter, and extract complex datasets.
- **Optimized database schemas & queries** using joins, aggregations, and indexing for fast data retrieval.

Projects

MediVault – Patient Medical History System with Fingerprint Identification
Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy, PyTorch, TensorFlow/Keras, Gemini, WebSockets

- Developed a full-stack medical platform integrating **AI-based medical image analysis** with secure electronic patient records. Integrated **four deep learning models** for brain tumor, pneumonia, bone fracture, and stroke detection. Built **FastAPI REST APIs** with PostgreSQL and SQLAlchemy for secure and efficient patient data management. Integrated **Google Gemini** for an AI-powered medical chatbot and document simplification, with **WebSocket-based real-time communication**. Implemented **fingerprint-based emergency patient identification**

Prompt Library — LLM-Powered Prompt Management & Generation System
Tech Stack: Python, FastAPI, Google Gemini API, JavaScript, REST APIs

- Developed an end-to-end LLM-powered Generative AI application managing 15 reusable prompt templates for Q&A, Summarization, and Creative Writing.
- Built FastAPI REST APIs for prompt retrieval, filtering, validation, and dynamic prompt construction.
- Integrated Google Gemini API with a JavaScript frontend for dynamic inputs and LLM-generated responses.

AI Customer Support Automation

- **Tech Stack:** n8n, LLM API, Telegram/WhatsApp, Google Sheets, Gmail, AI Agent/Memory
- Built an AI-powered customer support workflow using n8n, LLM API, and conversational memory for automated messaging across Telegram
- Automated customer responses, conversation logging, and data storage using Google Sheets, with Gmail integration for notifications.

Brain Tumor Classification Project

- **Tech Stack:** Python, Deep Learning, OpenCV, XceptionNet, Transfer Learning
- Built an end-to-end Computer Vision pipeline using **XceptionNet** to classify brain tumors from MRI scans.
- Applied **Transfer Learning** and **Data Augmentation** techniques to resolve class imbalance and boost classification accuracy.
- Evaluated model metrics using confusion matrices and classification reports to ensure high detection reliability.

AI-Powered Resume Screening

- **Tech Stack:** Python, NLP, Scikit-Learn, TF-IDF, Cosine Similarity
- Developed an NLP-based resume screening system to automatically rank candidates based on job descriptions.
- Applied **TF-IDF Vectorization** and **Cosine Similarity** for text feature extraction and relevance scoring.
- Trained and evaluated **Logistic Regression** and **Random Forest** models to automate candidate filtering.

Network Intrusion Detection System (NIDS)

- **Tech Stack:** Python, Machine Learning, Scikit-Learn, Streamlit, NSL-KDD
- Developed and deployed an end-to-end Machine Learning pipeline using **SVM** and **Random Forest** on the NSL-KDD dataset.
- Engineered data preprocessing workflows, incorporating log transformations and multi-strategy NaN handling (imputation/removal).
- Deployed an interactive **Streamlit** web application for real-time network threat prediction and log analysis.

Certificates

Data science and Machine Learning Diploma (Epsilon AI) , DIGITOPIA Hackathon - participant